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Patent results

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Use cross-encoder reranking

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Patent Intelligence

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Describe the invention or feature you are looking for

Cylindrical component surface treatment holder

Search patents

Results

Query: Cylindrical component surface treatment holder

Collection chunks

67,303

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2.02s

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Retrieved chunks

100

Unique patents

38

Returned patents

20

Reranked

Yes

rank	patent_number	title	score	evidence_chunks	max_similarity	top3_mean_similarity	max_rer
11	14904839	DEVICE FOR DISPENSING A DENTAL MATERIAL	-8.0119	5	0.0238	-0.0559	
12	14902717	POLYMERIC MATERIALS	-8.8859	1	-0.1703	-0.1703	
13	14904001	PLATED POLYMER COMPRESSOR	-8.8963	2	-0.1079	-0.1381	
14	14996264	LOCKING MECHANISM FOR A MEDICAL DEVICE ASSEMBLY TRAY	-8.9249	1	-0.1356	-0.1356	
15	14905328	STRUCTURE OF BATTERY PROTECTION CIRCUIT MODULE PACKAGE COUPLED WITH H	-8.987	1	-0.1248	-0.1248	
16	14904987	IMPLANTABLE DEVICE, IN PARTICULAR FOR CORRECTING AT LEAST ONE VERTEBRAL L	-9.2872	6	-0.1198	-0.137	
17	14988885	MONOLITHIC, GALLERYLESS PISTON AND METHOD OF CONSTRUCTION THEREOF	-9.3424	1	-0.1561	-0.1561	
18	14995875	LIQUID EJECTING HEAD AND LIQUID EJECTING APPARATUS	-9.4339	6	-0.0687	-0.0756	
19	14991470	NESTING COOKING VESSELS	-9.4762	1	-0.1154	-0.1154	
20	14993757	FLUID INPUT MODULE FOR MULTI-LUMEN CATHETERS	-9.7211	10	-0.1065	-0.1158	

1. 14904647 — METHODS AND APPARATUS FOR TREATING PULMONARY EMBOLISM

Patent score

-0.4760

Max similarity

0.0741

Top-3 mean

0.0451

Evidence chunks

11

Decision

REJECTED

Max reranker score

0.6236

Top-3 mean reranker score

-1.5756

Retrieved 11 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 71 · embedding rank 4 · similarity 0.0501 · reranker rank 1 · score 0.6236

##ot treatment device 1202, the device will have an external surface area between 1. 5× and 6× the surface area of a uniformly cylindrical device of the same general diameter of the cylindrical flow restoration portions 1212. in other preferred embodiments the ratio can be 2× to 4×. this can be advantageous particularly during retraction of the clot treatment device 1202 through the clot e. as shown in fig. 14f, the clot treatment device 1202 may become elongated as it is being withdrawn through the clot e. such elongation causes the clot material to encounter greater surface area of the clot treatment device 1202 than would otherwise occur with a device that was only generally cylindrical, e. g., that did not incorporate radially extending portions 1206. accordingly the clot treatment device 1202 is particularly adept at capturing the maximum amount of clot material during withdrawal. the clot treatment device 1202 is intended for use in large vessels, i. e., vessels with a diameter greater than 8 mm. for example, the diameter of the pulmonary arteries typically range from 15 to 30 mm whereas the first branches of the pulmonary

Chunk 63 · embedding rank 2 · similarity 0.0741 · reranker rank 4 · score -2.5812

shape with a linear (non - rounded) triangular cross - section. referring to fig. 17e, some of the capture portions 1206 between the generally cylindrical flow restoration portions 1212 of the clot treatment device 1202 are defined by a cylindrical disk shape with a rounded cross - section and others have a rectangular cross section. referring to fig. 17f, the radially extending portions 1206 between the generally cylindrical flow restoration portions 1212 of the clot treatment device 1202 alternate between cylindrical disk shape with a t - shaped cross - section and a flare - shaped cross - section. referring to fig. 17g, the radially extending portions 1206 between the generally cylindrical flow restoration portions 1212 of the clot treatment device 1202 are defined by a partial cylindrical disk shapes. referring to fig. 17h, the radially extending portions 1206 between the generally cylindrical flow restoration portions 1212 of the clot treatment device 1202 are defined by tabs and bumps or protuberances arising from the cylindrical surface of the device 1202. fig. 18 is a cross - sectional view of another embodiment of the clot treatment device 1202 in accordance with the technology

Chunk 47 · embedding rank 21 · similarity -0.0972 · reranker rank 5 · score -2.7691

atraumatic member (not shown) that extends distally from the device 1202. in yet further embodiments, the distal hub 1205 can be tapered to better penetrate the clot material in the vessel. in some embodiments, the clot treatment device 1202 can have a generally cylindrical shape that, during use, provides a flow lumen for blood across a clot. the treatment device 1202 is not, however, limited to a generally cylindrical shape. for example, the shape can be generally conical, generally concave or generally convex along its axis, so long as such shapes provide the aforesaid lumen for blood flow. referring still to figs. 12 and 13, the clot treatment device 1202 is compressed to fit within the diameter dl of a lumen 1407 of the delivery catheter 1406 in the undeployed state. in the deployed state shown in fig. 13, the clot treatment device 1202 has a plurality of capture elements, such as a series of radially extending capture portions 1206 which are separated from each other by flow restoration portions 1212. the flow restoration portions 1212 are configured

➤ 2. 14904285 — SUPPORTING TOOL FOR SUPPORTING CYLINDRICAL PARTS, LIKE LANDING GEAR RODS AND CYLINDERS

➤ 3. 14997983 — CHOCOLATE PROCESSING SYSTEM AND METHOD

▼ 4. 14992383 — BIOPSY SPACER DEVICE AND METHOD OF OPERATION

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
-3.6331	-0.1559	-0.1559	1	PENDING
Max reranker score	Top-3 mean reranker score			
-3.6331	-3.6331			

Retrieved 1 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 13 · embedding rank 70 · similarity -0.1559 · reranker rank 9 · score -3.6331

end piece 57 functions to limit the rod 61 and head piece 59 from being removed from the holder member 63 while providing for opening of the valve 47 for drainage. in an embodiment, the end piece 57 has a diameter of 0. 60 cm with a 0. 20 cm diameter circular recess sized to receive the rod 61. as will be discussed in more detail herein, the end piece 57 cooperates with a drainage device 100 (fig. 18) to open the valve 47. in an embodiment, the end piece 57 has a width of 0. 10 cm. the rod 61 connects the head piece 59 and end piece 57. in the exemplary embodiment, the rod 61 has a cylindrical shape with a diameter of 0. 20 cm and a height of 0. 50 cm. in operation, the connection of the rod 61 to the head piece 59 and end piece 57 allow components to move between open and closed position when assembled through the holder member 63. in one embodiment, the catheter 50 may have a manifold 60 (fig. 8) that bifurcates the catheter to allow multiple internal conduits 54, 56, 58 to be

▼ 5. 14904113 — LINEAR COMPRESSOR FOR A DOMESTIC APPLIANCE AND DOMESTIC REFRIGERATION APPLIANCE

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
-4.5331	-0.0108	-0.1019	3	PENDING
Max reranker score	Top-3 mean reranker score			
-2.9487	-6.1175			

Retrieved 3 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 7 · embedding rank 8 · similarity -0.0108 · reranker rank 7 · score -2.9487

of which the manufacture of the cylinder can be significantly facilitated, as a corresponding surface treatment need be performed only on the bushing and not on the entire cylinder. in a further advantageous embodiment of the invention it is provided that the respective predefined areas are embodied in steel with the alloy x15t. n. a particularly abrasion - resistant and low - wear surface is thereby provided on the areas predefined in each case. in a further advantageous embodiment of the invention it is provided that the microstructures of the areas predefined in each case comprise homogeneously distributed, spherical carbides. a particularly high degree of abrasion - resistance of the areas predefined in each case is thereby achieved. the term spherical should be taken to mean that the carbides are essentially embodied in spherical form, that is, within the usual manufacturing tolerances, have a maximum deviation from a theoretically optimal spherical shape in the range of about maximum 10 to 15 %. a further advantageous embodiment of the invention provides that the carbides have a diameter of 1 to 2 µm. a particularly fine microst

Chunk 8 · embedding rank 53 · similarity -0.1418 · reranker rank 17 · score -7.1935

range of about maximum 10 to 15 %. a further advantageous embodiment of the invention provides that the carbides have a diameter of 1 to 2 µm. a particularly fine microstructure in the form of ultra - finely distributed spherical carbides is thereby enabled, by means of which the abrasion - resistance of the predefined areas is further improved. according to a further advantageous embodiment of the invention it is provided that the areas predefined in each case are heat - treated, surface - hardened and / or press - hardened. by means of an appropriate setting of respective process parameters and monitoring of these set process parameters, a corresponding surface layer can be embodied in the areas predefined in each case, which essentially exhibit no porous surface, by means of which the desired wear - resistant and tough layer can be embodied on the respective surfaces. the corresponding microstructure can furthermore be embodied in a particularly simple manner by means of heat - treatment, surface hardening and / or press hardening. the inventive household refrigeration appliance with a refrigeration circuit and a compressor is characterized in that

Chunk 6 · embedding rank 65 · similarity -0.1532 · reranker rank 23 · score -8.2105

the predefined areas, where contact between the piston outer surface and the cylinder inner surface is possible if no gas pressure bearing is embodied or the gas pressure bearing has fallen below a predetermined pressure threshold value. the areas predefined in each case are thus only embodied with the martensitic

microstructure where the danger actually exists that in the case of an insufficient supply of pressure for the linear compressor, contact between the piston outer surface and the cylinder inner surface is at all possible. it is thereby possible to save a laborious complete processing of the piston outer surface and the cylinder inner surface, which has a positive effect on the manufacturing costs of the respective components. in a further advantageous embodiment of the invention it is provided for the cylinder to comprise a bushing, which comprises the cylinder inner surface. the correspondingly predefined area with the martensitic microstructure on the cylinder need only be embodied on the bushing, by means of which the manufacture of the cylinder can be significantly facilitated, as a corresponding surface treatment need be performed only on the bushing and not on the entire cylinder. in a further advantageous embodiment

6. 14905370 — CLUTCH RELEASE BEARING DEVICE

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
-6.2481	-0.1569	-0.1569	1	PENDING
Max reranker score			Top-3 mean reranker score	
-6.2481			-6.2481	

Retrieved 1 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 7 · embedding rank 73 · similarity -0.1569 · reranker rank 11 · score -6.2481

cylindrical portion is configured to be press - fitted to an inner periphery of the inner ring, wherein the protruding portion comprises an end portion having a convex circular - arc shape in cross - section, and has an outer diameter that is larger than an outer diameter of the cylindrical portion, and wherein a contact position between the protruding portion and the diaphragm spring is arranged between an extension line of an inner diameter of the inner ring and an extension line of an outer diameter of the inner ring. the contact position between the contact member and the diaphragm spring is located between the extension line of the inner diameter of the inner ring and the extension line of the outer diameter of the inner ring. with this, a force of inclining the inner ring is not applied, and hence a state of the contact with the diaphragm spring is stabilized. as a result, an unnecessarily great force is not applied to the sealing device, and hence stable sealing performance is maintained. further, the contact member is configured to be held in contact with the diaphragm spring at the protruding portion having the convex circular - arc shape in cross - section. with this, a straight line

7. 14903825 — METHOD FOR PREPARING A COATING OF A CYLINDRICAL SURFACE OF A SCANNING SLEEVE FOR A PRINTING MACHINE AND SCANNING SLEEVE THAT IS PREPARED BY THIS METHOD

8. 14993478 — INTRAVASCULAR CATHETER HAVING AN EXPANDABLE PORTION

Patent score	Max similarity	Top-3 mean	Evidence chunks	Decision
-7.0711	-0.1507	-0.1632	2	PENDING
Max reranker score			Top-3 mean reranker score	
-6.9280			-7.2142	

Retrieved 2 chunk(s) for this patent; showing the top 3 evidence chunks.

Chunk 8 · embedding rank 61 · similarity -0.1507 · reranker rank 13 · score -6.9280

blood vessel and expanded to create a stable platform. this stable platform can be configured to receive a plurality of medical treatment devices that may be manipulated by operating the catheter device. although the present invention will be described in the context of a number of embodiments, some of which may have particular medical applications, it should be appreciated that the present invention can be used in any desired environment and for any desired purpose. the illustrated catheter device 10 includes a handle assembly, indicated generally at 20. the illustrated handle assembly 20 includes an elongated, cylindrical handle body 21. the handle body 21 may alternatively have any other shape that is suitable for easy handling by a surgeon. further, the handle body 21 can be made from any suitably rigid material including, but not limited to, stainless steel or polymers. the handle assembly 20 is configured to selectively control the expansion and contraction of the expandable portion 90 by operation of a control mechanism 25. in the present embodiment, the control mechanism is shown as a slide that operates in slot 24. the slide permits the surgeon to operate the control mechanism 25 along the slot 24 in the handle body 21. this mechanism will be described in greater

Chunk 4 · embedding rank 99 · similarity -0.1758 · reranker rank 18 · score -7.5004

portion has a first diameter, and an opened position, wherein the expandable portion has a second diameter that is larger than the first diameter. an incising element is provided on the outer surface of at least one of the struts. the incising element has a blade that extends outwardly in a radial direction from the outer surface of the strut for creating an incision in atherosclerotic material located within a blood vessel when the expandable portion is in the opened position. the present invention additionally relates to a catheter platform device for use during endovascular surgery. the expandable portion is configured to provide a stable, flexible platform for medical treatment. the distal end of the catheter, including the expandable portion and the surrounding non - expandable portion, is capable of receiving a plurality of various medical treatment tools. brief description of the drawings in addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein : fig. 1 is a plan view of another exemplary

9. 14905117 — DENTAL IMPLANT WITH CORONAL GROOVE STRUCTURE

10. 14992023 — CONTROL NUCLEIC ACIDS FOR MULTIPLE PARAMETERS

11. 14904839 — DEVICE FOR DISPENSING A DENTAL MATERIAL

> 12. 14902717 — POLYMERIC MATERIALS

> 13. 14904001 — PLATED POLYMER COMPRESSOR

> 14. 14996264 — LOCKING MECHANISM FOR A MEDICAL DEVICE ASSEMBLY TRAY

> 15. 14905328 — STRUCTURE OF BATTERY PROTECTION CIRCUIT MODULE PACKAGE COUPLED WITH HOLDER, AND BATTERY PACK HAVING SAME

> 16. 14904987 — IMPLANTABLE DEVICE, IN PARTICULAR FOR CORRECTING AT LEAST ONE VERTEBRAL LEVEL

> 17. 14988885 — MONOLITHIC, GALLERYLESS PISTON AND METHOD OF CONSTRUCTION THEREOF

> 18. 14995875 — LIQUID EJECTING HEAD AND LIQUID EJECTING APPARATUS

> 19. 14991470 — NESTING COOKING VESSELS

> 20. 14993757 — FLUID INPUT MODULE FOR MULTI-LUMEN CATHETERS

▼ Pipeline inspection

Collection: patent_rag_chunks

Index path: /home/patzer_adi/Documents/python/
Python_Workspace/Patent_Intelligence_Project/data/
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Embedding model: sentence-transformers/all-
MiniLM-L6-v2

Reranker model: cross-encoder/ms-marco-MiniLM-L-6-v2

Stage timings

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  "vector_search" : 2.0218988950073253
  "reranking" : 3.4542227070051013
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Top retrieved chunks

embedding_rank	patent_number	chunk_index	embedding_similarity	reranker_rank	reranker_score
4	14904647	71	0.0501	1	0.6236
12	14904285	0	-0.0486	2	-0.3733
3	14997983	83	0.0518	3	-2.1663
2	14904647	63	0.0741	4	-2.5812
21	14904647	47	-0.0972	5	-2.7691
7	14904647	70	-0.0093	6	-2.8473
8	14904113	7	-0.0108	7	-2.9487
6	14904647	62	0.0109	8	-3.5443
70	14992383	13	-0.1559	9	-3.6331
63	14904647	50	-0.1518	10	-5.6346